

Modeling Chemical Kinetics with ChemKANs

Supervisor:

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Domain-Specific Lecture in Chemical Engineering:

- 166.242 [Chemical Process Engineering](#) (recommended)

Description:

Chemical kinetics describe how species concentrations evolve in time during reactive processes such as combustion or catalysis. Solving these detailed kinetic models often involves thousands of stiff coupled differential equations and is computationally demanding.

ChemKAN (Chemical Kolmogorov–Arnold Network) is a recently proposed neural network architecture that embeds chemical structure and physical constraints into a Neural ODE framework, enabling data-driven yet interpretable modeling of reaction dynamics (Koenig et al., Phys. Chem. Chem. Phys., 2025, DOI: 10.1039/D5CP02009C).

In this project, you will implement and train a ChemKAN model on literature-based chemical kinetics data, evaluate its predictive performance, and explore how well it captures underlying chemical dynamics compared to traditional solvers.

Your Task:

Perform an explorative data-driven study of chemical reaction dynamics using ChemKANs:

- Implement or adapt a ChemKAN architecture (e.g., in PyTorch).
- Train the model on time-series data from selected chemical systems (e.g., hydrogen or methane oxidation).
- Evaluate prediction accuracy, generalization, and computational efficiency.
- Visualize model behavior and interpret learned chemical dependencies.

Methods and Tools:

- Machine Learning (Neural ODEs, Kolmogorov–Arnold Networks),
- Data Analysis and Visualization,
- Python (PyTorch / TensorFlow),
- (Optional) Cantera or similar kinetics dataset generator.

Required Knowledge:

- Python, Machine Learning, Neural Networks,
- Basic understanding of ODEs or dynamical systems,
- Interest in scientific data modeling (chemical background not required).

